

Course Code	Name of the Course			
216U03E612	Instrumentation System			
Teaching Scheme	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	20	30	50
				100

Course prerequisites:

- Analog Electronics., Basic Electronics, Electrical Networks

Course Objectives:

Objectives of this course are:

- To understand the fundamental concepts of Electronics Measuring instruments.
- To explore the functioning and applications of various transducers such as temperature, displacement
- To learn Concepts of signal conditioning and data acquisition circuits
- To do case study for any feedback system through activity based learning.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand the working principle of various laboratory instruments
CO2. Choose a transducer for given application
CO3. Choose an appropriate signal conditioning circuits for instrumentation purpose
CO4. Rig up a circuit for measuring various physical parameters
CO5. Explore the working of PID controllers and PLC

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Introduction to Measurement and Instruments		08	CO1
	1.1	Fundamentals of Measurement: Units, Standards, and Measurement Errors		
	1.2	Electrical Measurement Instruments: Analog and Digital Instruments (Voltmeter, Ammeter, Wattmeter, Energy Meter)		
	1.3	Digital Instruments: Digital Storage Oscilloscopes (DSO), Arbitrary Function Generators, Spectrum Analyzers		
2	Sensors and Transducers		08	CO2
	2.1	Classification and Selection of Sensors and Transducers: Strain Gauges, Inductive & Capacitive Sensors, Piezoelectric, Hall Effect Transducers		
	2.2	Thermal Sensors (Thermistors, Thermocouples), Optical Sensors (Photodiodes, Phototransistors), Digital Transducers (Encoder Types)		
	2.3	Instrument Transformers: Current and Potential Transformers, Ratio and Phase Angle Errors		
3	Signal Conditioning Circuits		08	CO3
	3.1	Instrumentation Amplifiers, Precision Rectifiers, Isolation Amplifiers, Log Amplifiers, Charge Amplifiers		
	3.2	Noise Reduction Techniques in Instrumentation, Basics of Smart Sensors, Data Acquisition Systems (DAQ), Signal Conditioning and Telemetry		
4	Measurement of various Elements		07	CO4
	4.1	Displacement Measurement: Resistive, Capacitive, and Inductive Sensors, Differential Pressure Measurement		
	4.2	Torque Measurement (Strain Gauge Methods), Flow Measurement (Hotwire Anemometer, Eddy Current Sensors)		
	4.3	Phase Measurement: Analog and Digital Phase Detectors		
5	Process Control And Automation		14	CO5
	5.1	Basic Control Actions: ON-OFF, Proportional, Integral, Derivative Controls; PID Controllers; Controller Tuning Techniques (Ziegler-Nichols, Process Reaction Curve, etc.)		
	5.2	Programmable logic controllers, PLC architecture, programming, Relay ladder logic, Introduction to Sequence Control, PLCs - Working, Specifications, concept of latching, Introduction to Timer/Counters, Exercises based on Timers, Counters.		
	5.3	Advanced Process Automation: Introduction to SCADA, DCS, and IIoT Applications in Instrumentation		
Total			45	

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	A. K. Sawhney	<i>Electrical and Electronics Measurements and Instrumentation</i>	Dhanpat Rai and son's Ltd, New Delhi	2 nd Edition, 2003
2	DVS Murthy	<i>Transducers and Instrumentation</i>	PHI, Learning Pvt Ltd New Delhi	2 nd Edition, 2012
3	Curtis D Johnson	<i>Process Control Instrumentation Technology</i>	PHI, Learning Pvt Ltd New Delhi	8 th Edition, 2013
4	Doebelin E.O	<i>Measurement Systems: Application and Design</i>	McGraw Hill, Newyork,	4 th Edition, 1992
5	Madhuchhanda Mitra, Samarjit Sengupta	<i>Programmable Logic Controllers and Industrial Automation An Introduction</i>	Penram International Publishing (India) Pvt Ltd.,	2 nd Edition, 2017
6	S. Nihtianov, A. Luque	<i>Smart Sensors and MEMS: Intelligent Sensing Devices and Microsystems</i>	Woodhead Publishing	2nd Edition, 2018